SAMPA

USER GUIDE | Windows & Linux

Segment Anything Model for Plant Image Analysis

Compiled application • MATLAB R2025b • Version 1.0.0

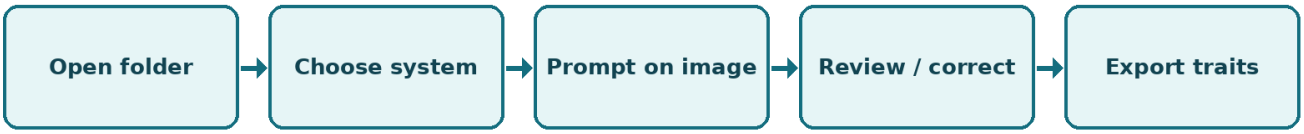


Figure 1. Five-step SAMPA analysis workflow.

At a glance

SAMPA lets users select plant structures from an input image using point, box, or polygon interactions; inspect distinct colour-coded labels; correct masks; and obtain per-label and combined root/shoot traits. This manual describes the compiled Windows and Linux versions; MATLAB source code is not needed by end users.

1. Before you install

Requirement	Guidance
System	64-bit Windows or Linux; use the package built for your operating system.
Runtime	Install the MATLAB Runtime matching MATLAB R2025b. No full MATLAB installation is required for end users.
Memory	Minimum 8 GB RAM; 16 GB recommended.
Processor	64-bit Intel/AMD; 4-core processor recommended.
GPU	Optional CPU execution; NVIDIA CUDA-capable GPU with at least 6 GB VRAM recommended for responsive segmentation, if supported by this compiled build.
Storage	Allow space for the installer, MATLAB Runtime, model dependencies, and images.

Check with the release maintainer whether GPU acceleration and standalone CPU operation have been tested for your distributed build. The current manual does not assume a particular checkpoint is bundled.

2. Install SAMPA on Windows

1. Download the Windows package from GitHub Release. Choose the release matching this manual.
2. Install MATLAB Runtime R2025b if it is not already installed. Use the official MathWorks Runtime installer; keep its default components unless your administrator specifies otherwise.
3. Run the supplied SAMPA Windows installer (SAMPA_tool.exe), then follow the installation wizard.
4. Launch SAMPA from its installed shortcut or application folder. Allow additional time for the first model initialization.
5. Open the supplied example images and confirm that one segmentation and trait export work.

If Windows shows a security prompt, verify the publisher and release source before choosing to run the application. Do not bypass institutional security policies.

3. Install SAMPA on Linux

6. Download the Linux distribution from GitHub Release and extract it into a directory you can access.
7. Install the MATLAB Runtime R2025b compatible with your Linux distribution.
8. Follow the included launcher instructions. For a MATLAB Compiler package this is typically a shell launcher that accepts the MATLAB Runtime installation directory; exact launcher name and arguments must be taken from the release README. The command for execution of Linux version is show below:

   -run_SAMPA_tool.sh (shell script for temporarily setting environment variables and executing the application). To run the shell script, type
   ./run_SAMPA_tool.sh <mcr_directory>
   at Linux or Mac command prompt.
   <mcr_directory> is the directory where MATLAB Runtime(R2025b) is installed or the directory where MATLAB is installed on the machine.
   Eg: ./run_SAMPA_tool.sh /mathworks/home/application/R2025b

9. If the launcher lacks execute permission, use chmod +x on that launcher file, provided you trust the download.
10. Run the application in a graphical desktop session and test it with the example images.

Linux note: MATLAB Runtime paths, library dependencies, and graphics/GPU drivers depend on the distribution. Follow the release-specific instructions rather than guessing a command or changing system library paths globally.

4. Identify the interface

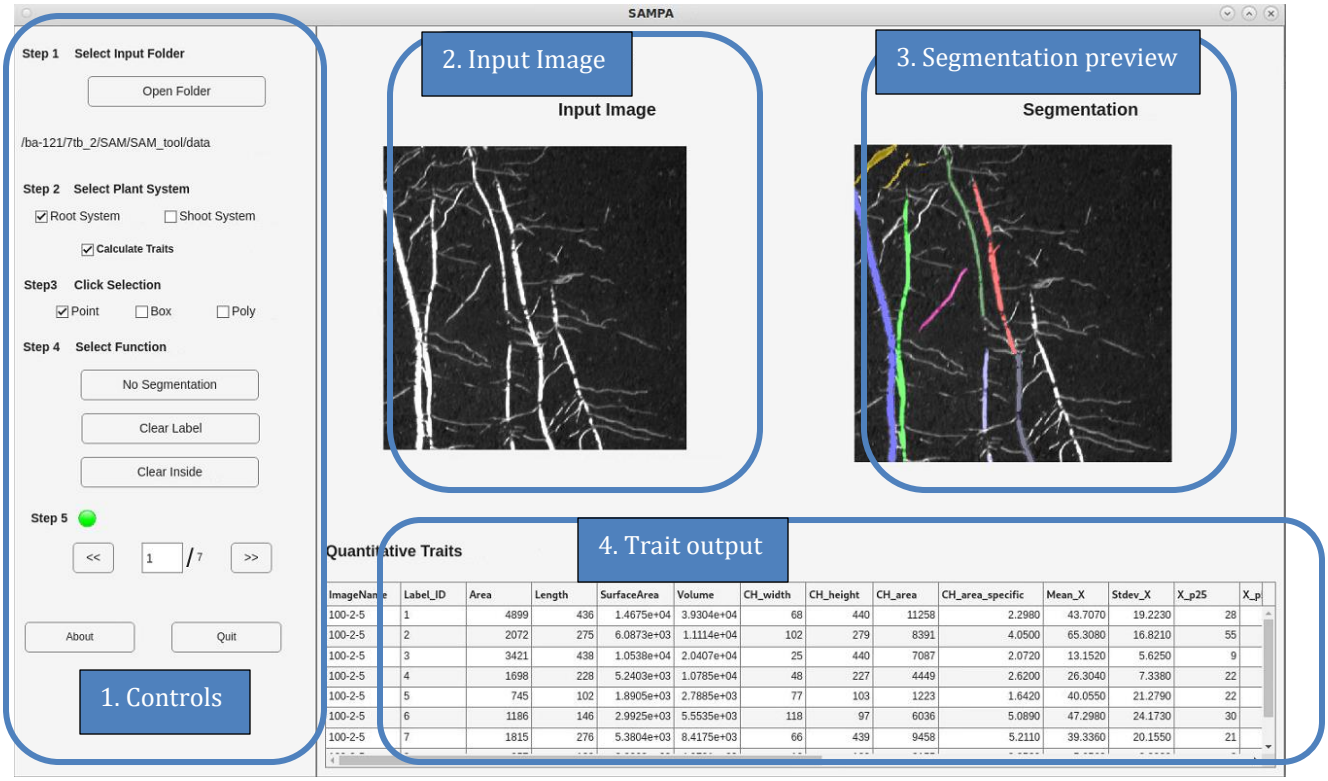


Figure 2. Annotated SAMPA interface (illustrative screenshot).

Number	Area	What to do
1	Controls	Open folder; choose root/shoot mode; select Point, Box or Poly; use correction and navigation buttons.
2	Input Image	Make every click, box selection, or polygon selection here.
3	Segmentation preview	Inspect the coloured masks and their independent label identities.
4	Quantitative Traits	Review one row per label and the “all” aggregate row.

5. Process an image

11. Click Open Folder and choose the folder containing your JPG, PNG, or TIFF root/shoot images.
12. Select Root System or Shoot System to choose the appropriate biological trait definitions.
13. Choose Point, Box, or Poly under the prompt controls.
14. Interact with the target structure in the Input Image panel (not the Segmentation preview).

15. Check the new coloured label in the preview, then correct it if needed.
16. Enable/click the trait-calculation control; review the per-label results and “all” row.
17. Use the previous/next image controls to continue through the folder.

6. Prompt modes and corrections

Control	Behaviour
Point	Click a foreground point on the desired organ; the red marker indicates the selected position.
Box	Draw a rectangle enclosing the object and confirm the box selection.
Poly	Draw a polygon. SAMPA prompts SAM with the polygon’s enclosing box, then sets predicted pixels outside the polygon to background.
Clear Label	Enter/select the Label ID to remove its mask while retaining other labels.
Clear Inside	Draw a polygon over unwanted interior or over-segmented pixels; remove that region from the current segmentation.
No Segmentation	Use to mark an image with no root/shoot target as unsegmented and skip the object. It is not whole-image foreground segmentation.

7. Understanding the outputs

SAMPA saves output automatically in a sibling output folder named after the input directory, for example dataImgs_out for dataImgs. Actual file availability depends on the packaged release.

Folder/file	Purpose
RootTraits.csv / ShootTraits.csv	Consolidated table of measured traits.
masks/*_mask.png	Binary mask per image.
masks/*_instance.tif	16-bit label/instance map retaining distinct Label IDs.
masks/*_overlay.png	Coloured segmentation overlay.
masks/*_CH.png	Convex-hull image.
prompts/*_point.csv, *_box.csv, *_polygons.csv	Recorded prompt geometry.
label_traits/*_RootTraits.csv or *_ShootTraits.csv	Per-image label-wise phenotypic traits.
log.txt	Processing and prompt history.

The “all” row refers to the union of currently labelled plant pixels. It is not necessarily the entire image rectangle.

8. Key phenotypic measurements

Root System	Shoot System
Area; skeleton-derived length; estimated surface area and volume	Projected shoot area; bounding-box height/width and area
Convex-hull width, height and area	Convex-hull area; shoot area / bounding-box area; shoot area / convex-hull area
X/Y coordinate statistics and percentile positions (e.g. X_p25 to X_p99)	Shape descriptors and colour statistics (RGB/HSV/Lab), where enabled

Root volume and surface area are image-derived cylindrical estimates rather than direct 3D measurements. Pixel-to-millimetre calibration should be handled consistently before comparing physical units.

9. Quick troubleshooting

Symptom	Suggested check
Application does not launch	Confirm operating system/package match, MATLAB Runtime R2025b installation, and release launcher instructions.
Model does not load	Check that model/support-package files shipped with the release are intact and accessible.
Slow segmentation	Try smaller images, fewer concurrent processes, or a supported GPU; CPU execution can be slower.
Unexpected missing roots	Use alternative prompts and inspect fine/low-contrast structures.
Wrong object or extra pixels	Try a tighter box/polygon, Clear Inside, or Clear Label followed by a new prompt.
Trait CSV appears in one column	Import as comma-delimited CSV or use a spreadsheet import wizard with comma delimiter.
Cannot find results	Check the automatically created input-folder-name_out directory.

10. Reproducibility and support

For repeatable experiments, retain the original images, image resolution and scale, SAMPA release version, prompt files, saved instance masks, trait CSVs and log.txt. Record whether segmentation was corrected manually and whether CPU or GPU inference was used.

Software release: [GitHub URL]

Support: narisetti@ipk-gatersleben.de | License: MIT License

Release checklist: verify executable filenames, Linux launcher syntax, exact model checkpoint, CPU/GPU compatibility, and installation screenshots against the final packaged version before publication.